

STATE BOARD OF TECHNICAL EDUCATION AND TRAINING
TELANGANA
DIPLOMA EXAMINATION (C-24)
C-24-SEP-2025
SEMESTER III, MID-I EXAM



EC-302
Digital Electronics

PCODE
243034

Exam Date: 02-09-2025

Session: AN

Duration: 1 Hour [01:30 PM To 02:30 PM]

[Total Marks: 20]

PART-A

Instructions:

1. Answer the following questions.
2. Each question carries **ONE** mark.

4 X 1 = 4

1. Mention applications of gray code.
2. State sign bit numbers for positive, negative numbers in binary system.
3. Write the Truth tables of universal logic gates
4. How many NAND & NOR gates are using to realize EX-OR gate.

PART-B

Instructions:

1. Answer the following questions.
2. Each question carries **THREE** marks.

2 X 3 = 6

- 5(a). Convert the octal number 714_8 to the decimal number.

— OR —

- 5(b). Convert Binary number 1100 to Excess-3 code.

- 6(a). Write the steps to convert SOP to Standard SOP form.

— OR —

- 6(b). Determine the output expression of SOP form for the given truth table and also draw logic diagram

Input			Output
A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	1

0	1	0	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

PART-C

Instructions:

1. Answer the following questions.

2 X 5 = 10

2. Each question carries FIVE marks.

7(a). Perform binary division for given numbers $01111100 \div 0010$.

— OR —

7(b). Subtract $(1010)_2$ from $(1111)_2$ using 1's complement method.

8(a). Realize EX-NOR Gate by using NAND & NOR gates.

— OR —

8(b). Realize AND, OR, NOT gates by using NAND gates.